Sahaj Bhadja

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Summary

Software engineer with 2+ years building automated test solutions, integration workflows, and production software across Linux environments. Experienced in Python, Java, C++, SQL, Git, CI/CD, object-oriented design, debugging, performance analysis, validation, and technical troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python services on Azure SQL and Blob Storage for 500+ users, optimizing queries, testable modules, and application performance to reduce production latency from 1.2 seconds to 350 milliseconds.
- Automated 5,000+ monthly integration jobs using explicit states, retries, validation, and failure isolation, tripling throughput while improving defect visibility, troubleshooting speed, maintainability, and reliable production execution.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask services and integration workflows handling 10,000+ daily requests, implementing status tracking, structured logging, validation, and backend test hooks while resolving production issues for 150+ users.
- Created automated processing for 1,000+ files with concurrency, AWS RDS persistence, executable validation checks, and structured failure handling, reducing test turnaround 60% while maintaining reliable application behavior.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible Linux-based simulation and validation pipelines processing 100,000+ records, adding automated test cases, runtime instrumentation, and systematic debugging to improve planning accuracy 18% across controlled evaluations.
- Executed repeatable integration experiments and performance benchmarks, documenting system configurations and troubleshooting complex system behavior while tuning an architecture that improved closed-loop stability 15% across simulated workloads.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, SQL integrations, and automated ETL tools using AWS Glue and S3, improving response times 30% and processing efficiency 68% across production and test environments.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating deep technical troubleshooting, regression validation, and production release confidence.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular Node.js services and WebSocket integrations across eight authentication, permissions, notifications, and operational workflows, synchronizing 2 million+ Azure SQL records through scalable, maintainable object-oriented service boundaries.
- Automated Docker builds, unit and integration tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving validation consistency and operational reliability.

Skills

Languages & Systems: Python, Java, C++, SQL, Bash, JavaScript, TypeScript, Linux, object-oriented programming, data structures, algorithms
Test Automation: PyTest, Mockito, unit testing, integration testing, regression testing, executable test cases, validation checks, test tooling, failure isolation, performance benchmarking
Data & Integration: PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB, AWS RDS, REST APIs, WebSockets, JSON, data modeling, query optimization, system integration
DevOps & Troubleshooting: Git, Docker, Kubernetes, Jenkins, CI/CD, AWS, Azure, scripting, debugging, logging, monitoring, root-cause analysis, production support, documentation, Agile

Projects

CPU-scheduling Simulator | Live | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with dynamic process inputs and Gantt visualization across 1,000+ combinations, reducing debugging time 20% through deterministic simulation, executable validation, and systematic edge-case testing.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MongoDB, Docker*
- Built a Linux-compatible application supporting 10+ databases, combining Java services, SQL validation, Docker, Maven, configuration checks, local debugging, and repeatable deployment workflows across Windows, macOS, and Linux.

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built containerized integration workflows supporting 50,000+ users and one million records, combining REST APIs, WebSockets, MongoDB, Kubernetes, monitoring, and automated CI/CD deployment for reliable production operation.